FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
FULL TEST – VIII
PAPER –2
TEST DATE: 05-05-2024

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. D

Sol. $4(3\cos t) - 3(4\cos t) + 2(5\sin t) = 5$

$\Rightarrow \sin t = 1/2$

$\therefore t = \frac{\pi}{6}\text{ sec}$

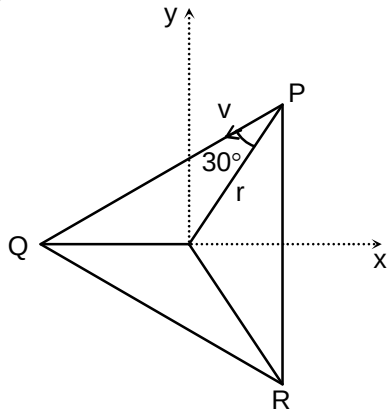
2. D

Sol. Along radial,

$$-\frac{dr}{dt} = \frac{\sqrt{3}}{2}v \quad \ldots(i)$$

Along tangent,

$$r\frac{d\theta}{dt} = \frac{v}{2} \quad \ldots(ii)$$



From (i) and (ii),

$$dr = -\sqrt{3}r\,d\theta$$

$$\int_{\ell/\sqrt{3}}^{r}\frac{dr}{r} = \int_{0}^{\theta} -\sqrt{3}d\theta$$

$$r = \frac{\ell}{\sqrt{3}}e^{-\sqrt{3}\theta}$$

Distance travelled when angualr dispalcement $d\theta$ is,

$$dS = \sqrt{(dr)^2 + (rd\theta)^2}$$

$$dS = \frac{2\ell}{\sqrt{3}}e^{-\sqrt{3}\theta}d\theta$$

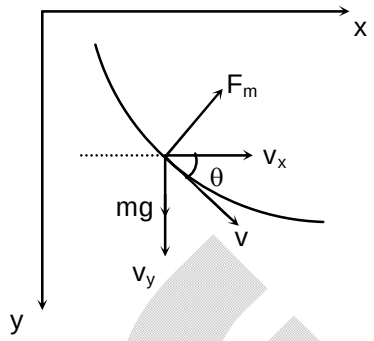
$$S = \int_{0}^{2\pi}\frac{2\ell}{\sqrt{3}}e^{-\sqrt{3}\theta}d\theta$$

$$S = \frac{2\ell}{\sqrt{3}}\left(1 - e^{-2\sqrt{3}\pi}\right)$$

3. A

Sol. For y-axis

$$mg - qBv_x = m\frac{dv_y}{dt} \quad \ldots(i)$$



For x-axis

$$qBv_y = m\frac{dv_x}{dt} \quad \ldots(ii)$$

Differentiating (i)

$$-qB\frac{dv_x}{dt} = m\frac{d^2v_y}{dt^2}$$

From (ii),

$$\frac{d^2v_y}{dt^2} = -\left(\frac{qB}{m}\right)^2 v_y$$

Solutio of above equation is

$v_y = v_{y_0} \sin\omega t$ v

At t = 0,

$$\frac{dv_y}{dt} = g$$

$$\therefore\ v_{y_0} = \frac{g}{\omega}$$

$$v_y = \frac{g}{\omega}\sin\omega t$$

From equation (i)

$mg - qBv_x = mg\cos\omega t$

$$v_x = \frac{mg}{qB}(1-\cos\omega t)$$

Since, at t = $\frac{\pi}{\omega}$, $v_y = 0$

$$v_x = \frac{2mg}{qB}$$

From work energy theorem,

$$mgy = \frac{1}{2}mv_x^2$$

$$\Rightarrow y = \frac{2m^2g}{q^2B^2} = 5 \text{ cm}$$

4. B

Sol. From constraint

$a_1 = 2a_2$

If $a_2 = a$ then $a_1 = 2a$

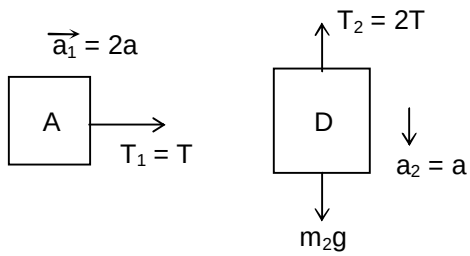
$T = m_1(2a)$ …(i)

$m_2g - 2T = m_2a$ …(ii)

Solving equation (i) and equation (ii)

$$a = \frac{5}{2} \text{ m/s}^2 \Rightarrow 2a = 5 \text{ m/s}^2$$

$T_1 = T = 75$ N



Speed of pulse, $v = \sqrt{\frac{T}{\mu}} = 50$ m/s

$$\frac{10 - \frac{5}{4}t^2}{50} = t$$

$t^2 + 40t - 8 = 0$

Solving we get, t = 0.2 sec

5. A, C, D

Sol. $mg\frac{L}{2} + mgL = \left(\frac{mL^2}{3} + mL^2\right)\alpha$

$\alpha = \frac{9g}{8L}$

Acceleration of ball is $\frac{9g}{8}$

For hinge force

$F_H = mg$

$2mg - F_v = 2m\frac{3L}{4}\cdot\frac{9g}{8}$

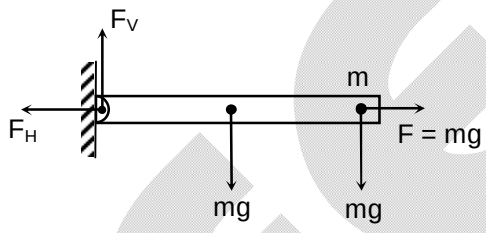
$F_v = \frac{5}{16}mg$

From work energy theorem

$mgL + mg\frac{L}{2} - mgL = \frac{1}{2}\left(\frac{mL^2}{3} + mL^2\right)\omega^2$

$\Rightarrow \omega = \sqrt{\frac{3g}{4L}}$

Speed of ball when the rod becomes vertical is $\sqrt{\frac{3gL}{4}}$



6. A, C, D

Sol. At t = 0, I = 40 μA

At $t \to \infty$, $I = \frac{48}{1.8\times10^6} = \frac{80}{3}\mu A$

Time count τ = 1 sec

Current as a function of time is

$I = \frac{80}{3}\left(1 - e^{-t}\right)\mu A$

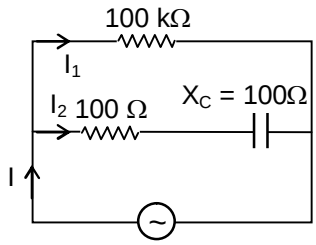
7. A, B

Sol. $I_1 = 2\sin\omega t$

$I_2 = \sqrt{2}\sin\left(\omega t + \frac{\pi}{4}\right)$

$I = I_1 + I_2$

$I = \sqrt{10}\sin\left(\omega t + \tan^{-1}\frac{1}{3}\right)$



E = 200 sin ωt Volt

Power factor $\cos\phi = \frac{3}{\sqrt{10}}$

Average power $P_{avg} = \left(\frac{1}{2}\right)(200)(\sqrt{10})\left(\frac{3}{\sqrt{10}}\right)$

$\therefore\ P_{avg} = 300\,\text{Watt}$

SECTION – B

8. 4

Sol. At the time of maximum compression

$Mv_0 = (m + 3m)v$

$v = \frac{v_0}{4}$, Here v is the speed of combined system

From conservation of energy;

$$\frac{1}{2}mv_0^2 = \frac{1}{2}4m\left(\frac{v_0}{4}\right)^2 + mgH + \frac{1}{2}k\left(\frac{H}{2}\right)^2$$

$\Rightarrow\ v_0 = \sqrt{4gH}$

9. 2

Sol. Time period is $T = 2\pi\sqrt{\frac{2L/5}{g}}$

$T = 2\pi\sqrt{\frac{2L}{5g}}$

$\therefore\ m = 2$

10. 3

Sol. $D = 1\,\text{mm} + 25 \times \frac{1}{100} = 1.25\,\text{mm}$

$$\left|\frac{\Delta Y}{Y}\right| \times 100 = \left(\frac{\Delta\ell}{\ell} + \frac{\Delta\varepsilon}{\varepsilon} + 2\frac{\Delta D}{D}\right) \times 100 = 2.41\%$$

11. 5

Sol. Resistance of uniform rod $R_1 = \frac{L}{k \cdot \pi R^2}$

Resistance of frustom $R_2 = \frac{L}{(2k)(\pi R)(2R)}$

$R_2 = \frac{L}{4k\pi R^2}$

$\therefore\ R_1 + R_2 = \frac{5L}{4k\pi R^2}$

Now if we consider the same lamina with equivalent thermal conductivity k_{eq}, then

$$R_{eq} = \frac{L}{k_{eq}\pi R^2} + \frac{L}{2k_{eq}\pi R^2} = \frac{3L}{2k_{eq}\pi R^2}$$

From (i) and (ii)

$k_{eq} = \dfrac{6k}{5}$

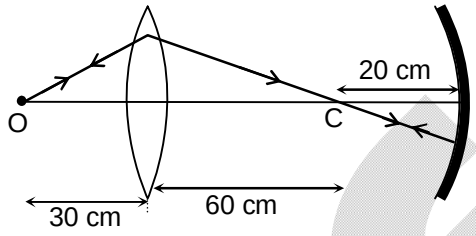
$\therefore \alpha = 5$

12. 200

Sol. 1st image formed by the lens must be at the centre of curvature of mirror or it must be at the pole of the mirror.

α = 110 cm

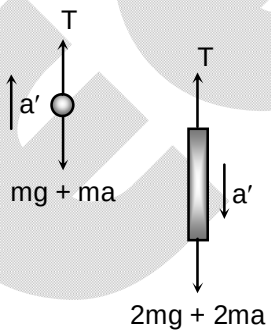
β = 90 cm



13. 35

Sol. For ball : $T - mg - ma = ma'$ …(i)

For rod : $2mg + 2ma - T = 2ma'$ …(ii)



Solving (i) and (ii); we get $a' = \dfrac{g+a}{3}$

$\therefore$ Relative acceleratio of rod with respect to ball is

$a_{rel} = 2a' = \dfrac{2}{3}(g+a)$

$\dfrac{dv_{rel}}{dt} = \dfrac{2}{3}(g+5t)$

$\Rightarrow \int_0^{v_{rel}} dv_{rel} = \int_0^t \dfrac{2}{3}(g+5t)$

$\Rightarrow v_{rel} = \dfrac{2}{3}\left(gt + \dfrac{5t^2}{2}\right)$

Again,

$\dfrac{dx_{rel}}{dt} = v_{rel} = \dfrac{2}{3}\left(gt + \dfrac{5t^2}{2}\right)$

$\Rightarrow \int_0^{\ell} dx_{rel} = \int_0^1 \dfrac{2}{3}\left(gt + \dfrac{5t^2}{2}\right)dt$

$\therefore \ell = \dfrac{35}{9}$ m

Here ℓ = length of the rod

SECTION – C

14. 8.00

Sol. Mass defect $\Delta m = \dfrac{Q}{C^2} = \dfrac{h^2}{24m\lambda^2C^2}$

$\therefore$ Mass of X is $16m + \dfrac{h^2}{24\, m\lambda^2C^2}$

$\Rightarrow \beta$ = 24 and α = 16

$\therefore \beta - \alpha$ = 8

15. 2.00

Sol. $X \longrightarrow Y + Z$

$$Q = K_Y + K_Z = \frac{h^2}{24m\lambda^2}$$

$$\frac{P^2}{24m} + \frac{P^2}{8m} = \frac{h^2}{24m\lambda^2}$$

$$\therefore\ P = \frac{h}{2\lambda}$$

De-Broglie wavelength of Y is 2λ

$\therefore$ n = 2

16. 260.00

Sol. ΔT can be found from the equilibrium condition.

$P_{right} = P_{left}$

$$\Rightarrow \frac{nR(T+\Delta T)}{A\left(\ell + \frac{\ell}{2}\right)} = \frac{nRT}{A\left(\ell - \frac{\ell}{2}\right)} + \frac{2k\left(\frac{\ell}{2}\right)}{A}$$

Solving we get

$$\Delta T = \frac{3k\ell^2}{2nR} + 2T = 260\ K$$

17. 2000.00

Sol. Let Q′ = Heat given to thermostat then from 1st law of thermodynamics

Q − Q′ = total work done by the gas in both the chambers + change in intenral energies of both the gases

$Q - Q' = W + \Delta U$

$$\Rightarrow Q - Q' = \frac{2k}{2}\left(\frac{\ell}{2}\right)^2 + \frac{3}{2}nR\Delta T$$

$$\Rightarrow Q - Q' = \frac{k\ell^2}{4} + \frac{3}{2}nR\left(\frac{3k\ell^2}{2nR} + 2T\right)$$

$$\Rightarrow Q' = Q - \frac{5k\ell^2}{2} - 3nRT$$

$\Rightarrow Q' = 12000 - 2500 - 7500$

$\therefore Q' = 2000\ J$

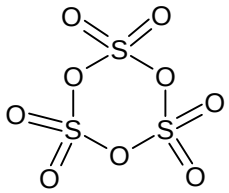
Chemistry

PART – II

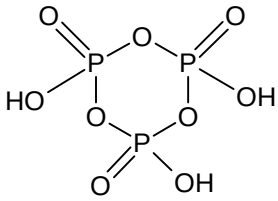
SECTION – A

18. C

Sol.



$\gamma - SO_3$



$(HPO_3)_3$

19. C

Sol. Aromatic.

20. B

Sol. Subsidiary quantum number 5 indicates h subshell wih 11 degenerate orbitals; the orbitals can take up a maximum of 22 electrons.

The maximum value will be given by 11 unpaired electrons.

$$S = \frac{1}{2} \times 11$$

$$\text{Spin multiplicity } = 2S + 1 = \frac{11}{2} \times 2 + 1 = 12$$

The minimum value will be given when no unpaired electron is present

S = 0

Spin multiplicity = 2S + 1

= 1

21. C

Sol. Thiols are more acidic than alcohols but weaker acid than carboxylic acid.

22. A, B, D

Sol. Heat of combustion $\propto$ number of carbon atoms.

Stability of carbocation.

Aromatic carbocation > allylic 2° carbocation > 2° carbocation > anti aromatic carbocation

Stabler the alkene lesser the heat of hydrogenation.

23. A, B

Sol. Fac-mer isomerism is shown by complexes of type $[MA_3B_3]$ and $[M(AB)_3]$.

24. B, D

Sol. $K_p = K_c (RT)^{\Delta n}$

$$\log K_p = \log K_c + \Delta n \log RT$$

$$\log\left(\frac{K_p}{K_c}\right) = \Delta n \log RT$$

$\therefore$ Slope $\Delta n = \tan\theta = 2$

SECTION – B

25. 213

Sol. $PH_3(g) \longrightarrow P(g) + 3H(g)$

$\Delta H^o_{atom} = 954 \text{ kJ mol}^{-1}$

$\therefore P - H_{\text{average BE}} = \dfrac{954}{3} = 318 \text{ kJ mol}^{-1}$

$P_2H_4(g) \longrightarrow 2P(g) + 4H(g) \qquad \Delta H^-_{atm} = 1485$

$\therefore P - P_{BE} + 4 \times P - H_{BE} = 1485$

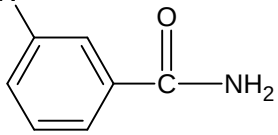
$P - P_{BE} + 4 \times 318 = 1485$

$\therefore P - P_{BE} = 1485 - 1272$

$= 213 \text{ kJ mol}^{-1}$

26. 40

Sol. H_2N



A

Molar mass of A = 136

$\therefore$ Millimoles of $A = \dfrac{2.72}{136} \times 1000$

= 20

1 mol of A $\equiv$ 1 mol of H_2SO_4

$\therefore$ Millimoles of $H_2SO_4 = 20$

$V \times 0.5 = 20$

V = 40 ml

27. 5

Sol. Mass of $CaCO_3$ = 10 g

Moles of $CaCO_3 = \dfrac{1}{100}$

Volume of CO_2 in air $V_{CO_2} = \dfrac{nRT}{P}$

$= \dfrac{10}{100} \times \dfrac{0.082 \times 300}{0.82}$

= 3

Volume fraction of CO_2 = 0.3

Solubility of CO_2 in water $= 3 \times 10^{-1} \times 3.333 \times 10^{-2}$

$= 10^{-2}$

Solubility of CO_2 will be initial concentration of H_2CO_3

$\left[H^+\right] = \sqrt{K_{a_1} \times C}$

$= \sqrt{10^{-8} \times 10^{-2}} = 10^{-5}$

pH = 5

28. 50

Sol. At high pressure

$$Z = 1 + \frac{Pb}{RT}$$

$$\text{Slope}\frac{b}{RT} = \frac{\pi}{49.26} \text{ atm}$$

$$b = \frac{\pi \times 0.0821 \times 300}{49.26}$$

$$= \frac{\pi}{2}$$

$$b = 4 \times N_A \times V$$

$$\frac{\pi}{2} = 4 \times 6 \times 10^{24} \quad \frac{4}{3} \times \pi r^3$$

$$r^3 = \frac{3}{2 \times 4 \times 4 \times 6 \times 10^{24}} \text{ dm}^3$$

$$r = \frac{1 \times 10^{-8}}{4} \text{dm}$$

$$= \frac{1}{4} \times 10^{-9} \text{ cm}$$

$$= 2.5 \times 10^{-9} \text{ cm}$$

= 25 pm

d = 50 pm

29. 3

Sol. Oxalic acid, malonic acid and picric acid are stronger than benzoic acid.

30. 5

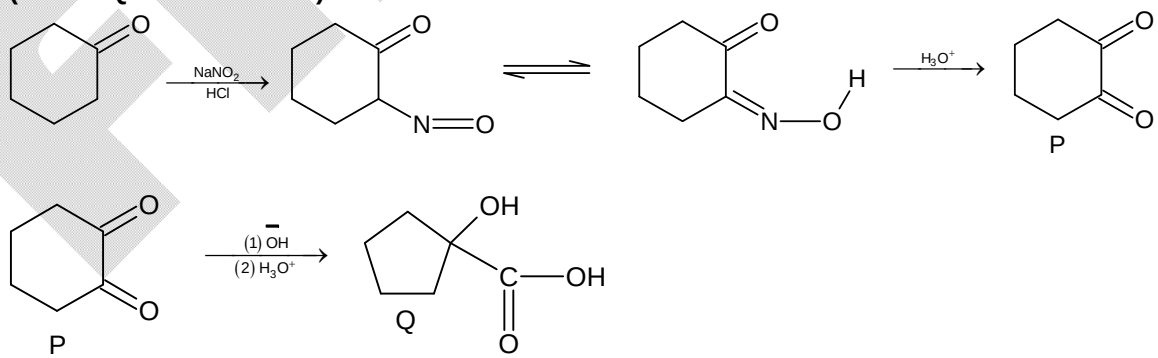
Sol. Xenon hexafluoroplatinate is the first noble gas compound.
He and Ne do not form clathrates with hydroquinone as they are very small.

SECTION – C

31. 64.29

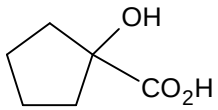
32. 6.50

Sol. **(for the Q. No. 31 and 32)**



Molar mass of P = 112

$$\% \text{ of Carbon in P} = \frac{72}{112} \times 100$$
$$= 64.285$$



Q

Molar mass of Q = 130

$$\therefore \frac{130}{20} = 6.5$$

33. 0.25

34. 0.25

Sol. **(for the Q. No. 33 and 34)**

In neutral or faintly alkaline medium

$$2MnO_4^- + H_2O + I^- \longrightarrow 2MnO_2 + 2OH^- + IO_3^-$$

Millimoles of $KMnO_4 = 500$

Millimoles of I^- oxidized = 250

Moles of I^- oxidized = 0.25

In acidic medium

$$10I^- + 2MnO_4^- + 16H^+ \longrightarrow 2Mn^{2+} + 8H_2O + 5I_2$$

Millimoles of $KMnO_4 = 50$

Millimoles of I^- oxidized = 250

Moles of I^- oxidized = 0.25

Mathematics

PART – III

SECTION – A

35. D

Sol. Number of ways = Coefficient of x^{30} in $[(x^{\circ} + x^2 + x^3) + x^4(x + 1)]^7 = 420$

36. D

Sol. Let $S = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\left[\frac{mn}{3^n} \times \frac{n\cdot 3^m + m\cdot 3^n - n\cdot 3^m}{3^m\left(n\cdot 3^m + m\cdot 3^n\right)}\right]$

$= \sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\left[\frac{m^2 n\cdot 3^n}{3^n\cdot 3^m\left(n\cdot 3^m + m\cdot 3^n\right)}\right] = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\left[\frac{m^2 n}{3^m\left(n\cdot 3^m + m\cdot 3^n\right)}\right]$

Interchanging m and n

$S = \sum_{n=1}^{\infty}\sum_{m=1}^{\infty}\frac{n^2 m}{3^n\left(m\cdot 3^n + n\cdot 3^m\right)}$

$\Rightarrow 2S = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\frac{mn}{3^m\cdot 3^n} = \left(\sum_{m=1}^{\infty}\frac{m}{3^m}\right)^2 = \frac{9}{16}$

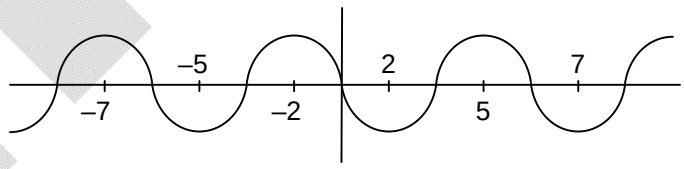
$\Rightarrow S = \frac{9}{32} = \frac{p}{q} \Rightarrow q - 3p = 32 - 27 = 5$

37. C

Sol. $\lim_{n\to\infty}\frac{\cos\frac{\pi}{2^2}\cos\frac{\pi}{2^3}.....\cos\frac{\pi}{2^{n-1}}}{\left(\cos\frac{\pi}{2^3}\cdot\cos\frac{\pi}{2^4}.....\cos\frac{\pi}{2^n}\right)^4}$

38. B

Sol. f is an odd function



39. B, D

Sol. Here, $\Delta'(x) = Ax + B$

$\Delta''(x) = A$

$\Delta'''(x) = 0$

40. A, D

Sol. $p'(x) = \lambda[(x - \beta)^2 + \gamma^2 + 2(x - \alpha)(x - \beta)]$

41. A, C, D

Sol. Given $\vec{a} = \vec{b}\times\vec{c} + 2\vec{b}$

$\Rightarrow |\vec{a}||\vec{b}|\cos\theta = 8$

$\Rightarrow \cos\theta = \frac{4}{|\vec{a}|} \Rightarrow |\vec{a}| = 4$ and $\theta = 0^{\circ}$

$\Rightarrow \vec{b}\times\vec{c} = 0 \Rightarrow$ Either $\vec{b} = \vec{c}$ or $\vec{b} = -\vec{c}$

$\Rightarrow 2\vec{a} + \vec{b} + \vec{c} = 2\vec{a}$ or $2\vec{a} + \vec{b} + \vec{c} = 2\vec{a} + 2\vec{b} = 3\vec{a}$

SECTION – B

42. 7

Sol. 111111, 888888 and numbers formed by (8, 8, 8, 1, 1, 1) are only the 6 digit numbers which are divisible by 21

$\Rightarrow n_3 = 20, n_2 = 11, n_1 = 2$

43. 5

Sol. $\int \frac{(e^x-1)\sin x-(e^x-1-x)\cos x}{1+(e^x-1-x-\cos x)(e^x-1-x+\cos x)}dx$

$= \int \frac{(e^x-1)\sin x-(e^x-1-x)\cos x}{\sin^2 x+(e^x-1-x)^2}dx$

$= \int \frac{(e^x-1)\sin x-(e^x-1-x)\cos x}{\sin^2 x\left[1+\left(\frac{e^x-1-x}{\sin x}\right)^2\right]}dx$

Let $\frac{e^x-1-x}{\sin x}=t$

$I = \tan^{-1}\left(\frac{e^x-1-x}{\sin x}\right)+c$

44. 8

Sol. Image of B in the crease always lies on line AD. Thus the parabola formed has the focus at B and directrix AD

45. 7

Sol. $x_1 = \frac{1}{2},\ x_2 = \frac{\sqrt{3}}{2}$

Now, $f'(x) = 1+x\left(2x+\frac{2}{3}\cdot 4x^3+\frac{2}{3}\cdot\frac{4}{5}\cdot 6x^5+.....\right) = 1+x\frac{d}{dx}(xf(x))$

$\Rightarrow (1-x^2)f'(x) = 1 + xf(x) \Rightarrow f(0) = 0$ and $f(x)=\frac{\sin^{-1}x}{\sqrt{1-x^2}}$

Now, area = $\int_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \frac{\sin^{-1}x}{\sqrt{1-x^2}}dx = \frac{\pi^2}{24} \Rightarrow a + b - 19 = 7$

46. 6

Sol. Let $z_1 = 1 \Rightarrow |1 + z_2| = |1 + \cos\theta + i\sin\theta|^2 = 2 + 2\cos\theta \geq 2+\sqrt{3}$

$\Rightarrow \cos\theta \geq \frac{\sqrt{3}}{2}$ or $\theta \in \left[-\frac{\pi}{6}, \frac{\pi}{6}\right]$ because $z_2 \neq 1$ and $\theta = \frac{2k\pi}{42}$

$\Rightarrow k = \pm 1, \pm 2, \pm 3 \Rightarrow P = \frac{6}{41}$

47. 8

Sol. $f\left(\frac{1}{x}\right)f'(x) = 1$ (Putting $x = \frac{1}{x}$)

$$\Rightarrow f''(x)f\left(\frac{1}{x}\right)+f'(x)f'\left(\frac{1}{x}\right)\left(-\frac{1}{x^2}\right)=0$$

$$\Rightarrow \frac{f''(x)}{f'(x)}-\frac{f'(x)}{f(x)}\cdot\frac{1}{x^2}=0$$

$$\Rightarrow \int\frac{f''(x)f(x)}{(f'(x))^2}dx=\int\frac{1}{x^2}dx$$

$$\Rightarrow \frac{-f(x)}{f'(x)}+x=-\frac{1}{x}+c$$

SECTION – C

48. 12.00

49. 2.00

Sol. (for Q. No. 48 and 49)

$x + 2y + a = 0$, $12x – 6y – 41 = 0$ and radical axis of the given circles will be concurrent

$\Rightarrow a = 2$

Now for required circle $S = 0$

$x^2 + y^2 – 4 + \lambda(x + 2y + 2) = 0$

(Equation of family of circles passing through the intersection points of $x + 2y + a=0$ and $x^2+y^2=4$)

Now, common chord with C_2 is $(\lambda + 4)x + 2(\lambda + 1)y + 2\lambda – 5 = 0$

Which is same as $12x – 6y – 41 = 0 \Rightarrow \lambda = -\frac{8}{5}$

50. 1.00

51. 3.00

Sol. $f(x + y + z) = f(x) f(y) f(z)$

Putting $y = z = –1$

$f(x – 2) = f(x) f(–1) f(–1)$

Putting $x = 2$

$f(0) = 4(f(–1))^2 \Rightarrow f(0)$ is positive

Now, putting $x = 0 = y = z$

$f(0) = f(0)^3 \Rightarrow f(0) = 1$

Again, putting $y = 2$ and $z = 0$

$f(x + 2) = f(x) f(2) f(0)$

$\Rightarrow f(x + 2) = 4f(x)$

$\Rightarrow f'(x + 2) = 4f'(x)$

$\Rightarrow f'(2) = 12$